

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0907

Dixon Landing 230 kV
January 15, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0907. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Dixon Landing 230 kV substation in the Ohio area. The request was received on December 17, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 251 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$181,647,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gould / Dixon Landing 230.0 kV
Contingency: Troy Energy Llc / Dixon Landing 230.0 kV
Loading: 228.5% (Rating: 656 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
2. Facility: Troy Energy Llc / Dixon Landng 230.0 kV
Contingency: Gould / Dixon Landing 230.0 kV L
Loading: 222.3% (Rating: 675 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
3. Facility: Trego 500 kV/1 kV Auto
Contingency: Troy Energy Llc Substation / Trego 500.0 kV
Loading: 138.6% (Rating: 936 MVA)
DFAX: 22.2%
Pre-project loading: 103.0%
4. Facility: Japanese Mesa / Brook Road 100.0 kV Line
Contingency: Dixon Landing / Troy Energy Llc Substation 230.0 kV Line
Loading: 134.1% (Rating: 136 MVA)
DFAX: 6.7%
Pre-project loading: 60.5%
5. Facility: Highgate / Trego 230.0 kV
Contingency: Troy Energy Llc / Trego Substation 500.0 kV
Loading: 130.6% (Rating: 822 MVA)
DFAX: 21.0%
Pre-project loading: 92.2%

6. Facility: Trego 230 kV/1 kV XFMR
Contingency: Trego / Troy Energy Llc 500.0 kV
Loading: 128.8% (Rating: 1,007 MVA)
DFAX: 22.2%
Pre-project loading: 95.8%
7. Facility: Highgate / Troy Energy Llc Substation 230.0 kV
Contingency: Trego / Troy Energy Llc 500.0 kV line
Loading: 128.1% (Rating: 1,214 MVA)
DFAX: 24.6%
Pre-project loading: 97.6%
8. Facility: Napoleonville / Brook Rd Substation 100.0 kV
Contingency: Dixon Landing / Troy Energy Llc 230.0 kV
Loading: 126.2% (Rating: 125 MVA)
DFAX: 6.7%
Pre-project loading: 46.3%
9. Facility: Napoleonville / Westland 1 Ra 100.0 kV
Contingency: Dixon Landing / Troy Energy Llc 230.0 kV line
Loading: 125.7% (Rating: 106 MVA)
DFAX: 6.7%
Pre-project loading: 31.3%
10. Facility: Imperial / Gruver 230.0 kV
Contingency: Imperial / Eufaula Substation 500.0 kV
Loading: 124.1% (Rating: 574 MVA)
DFAX: 7.1%
Pre-project loading: 105.6%
11. Facility: Joso / Eufaula 230.0 kV
Contingency: Beaver / Eufaula 500.0 kV
Loading: 121.8% (Rating: 783 MVA)
DFAX: 8.9%
Pre-project loading: 104.8%
12. Facility: Corridor / Larson St Substation 230.0 kV
Contingency: Trego / Beaver 500.0 kV
Loading: 116.9% (Rating: 1,093 MVA)
DFAX: 10.5%
Pre-project loading: 102.5%
13. Facility: Onancock Diesel Generation Facility / Joso 230.0 kV L
Contingency: Beaver / Eufaula 500.0 kV line
Loading: 116.7% (Rating: 2,010 MVA)
DFAX: 15.7%
Pre-project loading: 105.0%
14. Facility: Corridor / Lookingglass 230.0 kV
Contingency: Beaver / Trego 500.0 kV
Loading: 116.0% (Rating: 1,127 MVA)
DFAX: 10.5%
Pre-project loading: 102.0%
15. Facility: Truman / Lookingglass 230.0 kV
Contingency: Beaver / Trego Substation 500.0 kV line
Loading: 115.1% (Rating: 1,162 MVA)
DFAX: 10.5%
Pre-project loading: 101.6%
16. Facility: Myrtle Pt / Galion Generating Sta 230.0 kV
Contingency: Trego / Beaver 500.0 kV

- Loading: 114.5% (Rating: 1,298 MVA)
DFAX: 7.9%
Pre-project loading: 105.3%
17. Facility: Trumn Substation / Waldo 230.0 kV line
Contingency: Trego / Beaver 500.0 kV
Loading: 114.5% (Rating: 1,191 MVA)
DFAX: 10.5%
Pre-project loading: 101.2%
18. Facility: Baraga Vlg 500/230 kV Autotransformer
Contingency: Trego / Beaver 500.0 kV line
Loading: 114.3% (Rating: 1,515 MVA)
DFAX: 11.4%
Pre-project loading: 103.1%
19. Facility: Imperial / Chamberlin 230.0 kV
Contingency: Hylebos / Ravenna 500.0 kV Line
Loading: 113.2% (Rating: 1,299 MVA)
DFAX: 11.3%
Pre-project loading: 100.2%
20. Facility: Eufaula Substation 230 kV/1 kV Auto
Contingency: Eufaula / Imperial 500.0 kV line
Loading: 112.7% (Rating: 2,150 MVA)
DFAX: 12.2%
Pre-project loading: 104.2%
21. Facility: Euaula 500/1kV XFMR
Contingency: Imperial / Eufaula 500.0 kV
Loading: 112.7% (Rating: 2,150 MVA)
DFAX: 12.2%
Pre-project loading: 104.2%
22. Facility: Madeline / Baraga Vlg 230.0 kV
Contingency: Baraga Vlg / Gray'S Flat 230.0 kV line
Loading: 112.6% (Rating: 842 MVA)
DFAX: 5.4%
Pre-project loading: 103.0%
23. Facility: Waldo 500/230 kV Transformer Bank
Contingency: Trego / Beaver 500.0 kV
Loading: 111.5% (Rating: 1,310 MVA)
DFAX: 10.4%
Pre-project loading: 99.6%
24. Facility: James De Young Substation / Carley Rd 230.0 kV
Contingency: EMethuen 500-230 kV TX
Loading: 111.3% (Rating: 991 MVA)
DFAX: 5.8%
Pre-project loading: 102.4%
25. Facility: Baraga Vlg / Gray'S Flat 230.0 kV
Contingency: E Methuen / Madeline 230.0 kV
Loading: 111.2% (Rating: 1,355 MVA)
DFAX: 7.4%
Pre-project loading: 103.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gould / Dixon Landing 230.0 kV	Rebuild line with higher-capacity conductor	\$6,724,000	100% (\$6,724,000)
Dixon Landing / Troy Energy Llc 230.0 kV	Rebuild line with higher-capacity conductor	\$17,528,000	100% (\$17,528,000)
Trego 500/1kV Transformer Bank	Replace transformer with higher-rated unit	\$23,869,000	50% (\$11,888,000)
Brook Rd Substation / Japanese Mesa 100.0 kV	Rebuild line with higher-capacity conductor	\$4,007,000	92% (\$3,680,000)
Highgate / Trego 230.0 kV	Rebuild line with higher-capacity conductor	\$40,928,000	62% (\$25,402,000)
Trego 230 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$14,710,000	50% (\$7,327,000)
Highgate / Troy Energy Llc Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$42,907,000	59% (\$25,392,000)
Brook Rd / Napoleonville 100.0 kV	Rebuild line with higher-capacity conductor	\$2,010,000	92% (\$1,846,000)
Westland 1 Ra / Napoleonville 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,757,000	92% (\$2,531,000)
Imperial / Gruver 230.0 kV	Rebuild line with higher-capacity conductor	\$9,057,000	28% (\$2,570,000)
Joso / Eufaula Substation 230.0 kV	Rebuild line with higher-capacity conductor	\$11,260,000	27% (\$3,090,000)
Corridor / Larson St 230.0 kV Line	Rebuild line with higher-capacity conductor	\$11,271,000	26% (\$2,879,000)
Onancock Diesel Generation Facility / Joso 230.0 kV L	Rebuild line with higher-capacity conductor	\$19,551,000	26% (\$5,155,000)
Lookingglass / Corridor 230.0 kV Line	Rebuild line with higher-capacity conductor	\$24,250,000	26% (\$6,195,000)
Truman / Lookingglass 230.0 kV	Rebuild line with higher-capacity conductor	\$26,961,000	26% (\$6,887,000)
Myrtle Pt / Galion Generating Sta 230.0 kV	Rebuild line with higher-capacity conductor	\$43,870,000	23% (\$10,161,000)
Truman / Waldo 230.0 kV	Rebuild line with higher-capacity conductor	\$25,658,000	26% (\$6,554,000)
Baraga Vlg 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$20,031,000	20% (\$4,056,000)
Chamberlin / Imperial 230.0 kV	Rebuild line with higher-capacity conductor	\$10,818,000	23% (\$2,453,000)
Eufaula Substation 230 kV/1 kV Auto	Replace transformer with higher-rated unit	\$6,665,000	24% (\$1,586,000)
Eufaula Substation 500/1 kV XFMR	Replace transformer with higher-rated unit	\$32,930,000	24% (\$7,835,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Madeline / Baraga Vlg 230.0 kV	Rebuild line with higher-capacity conductor	\$13,445,000	19% (\$2,550,000)
Waldo 500 kV/230 kV Autotransformer	Replace transformer with higher-rated unit	\$26,114,000	29% (\$7,560,000)
James De Young Substation / Carley Rd 230.0 kV	Rebuild line with higher-capacity conductor	\$39,794,000	17% (\$6,937,000)
Baraga Vlg / Gray'S Flat 230.0 kV L	Rebuild line with higher-capacity conductor	\$14,816,000	19% (\$2,861,000)
	Total allocated to project		\$181,647,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.